AN IMPROVED LOWER BOUND FOR THE FURSTENBERG–SÁRKÖZY PROBLEM

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ABSTRACT. Let $D(N)$ be the largest size of a subset of $\{1,\dots,N\}$ no two of whose elements differ by a nonzero perfect square. We prove $\liminf_{N\to\infty} \log D(N)/\log N \geq \alpha_\infty = 0.753741541837329405\dots$, where α_∞ is the value of an explicit closed-form optimization over an eleven-block pool: nine of Krachun's Paley chains together with two square-DAG certificates on the square-free composite moduli 235 and 299. The previous record is Krachun's $\alpha_\star = 0.752796455874514\dots$, the first bound past 3/4. Only one lemma is new: a lift of square-DAGs on square-free composite moduli; Krachun's gluing lemmas are used as published. As a corollary, for prime q the point–line incidence graph of $\mathbb{F}_q^2$ contains an induced matching of size $\gg_\varepsilon q^{1/2+\alpha_\infty-\varepsilon}$. The full argument, including the numerical value, is formalized and machine-checked in Lean 4.

1. Introduction

By the Furstenberg–Sárközy theorem [2, 10], a set of integers of positive upper density contains two elements whose difference is a perfect square. Quantitatively, let $D(N)$ denote the largest size of a subset of $\{1,\dots,N\}$ no two of whose elements differ by a nonzero perfect square; such sets are called *square-difference-free* (SDF). The best upper bound is due to Green and Sawhney [3]: $D(N) \ll Ne^{-c\sqrt{\log N}}$. In the other direction, the history of the lower-bound exponent is short:

Year	Source	Exponent
1984	Ruzsa [9]	0.733077
2008, 2015	Beigel–Gasarch [1]; Lewko [7] (independently)	0.733412
2026	Krachun [5]	0.752796...
2026	this note	0.753741...

Theorem 1. $\liminf\limits_{N\to\infty} \dfrac{\log D(N)}{\log N} \geq \alpha_\infty = 0.753741541837329405\dots$ (nearest IEEE double 0.7537415418373294), *where α_∞ is given by the explicit closed form* (2) *evaluated on the eleven-block pool of Section 5: nine Paley chains and two square-DAG certificates on the composite moduli* 235 *and* 299, *lifted by Lemma A below, glued by Krachun's Lemma 5, and converted into integer sets by his Lemma 4.*

Equivalently, in pointwise form: for every $\varepsilon > 0$ there is an $N_0(\varepsilon)$ with $D(N) \geq N^{\alpha_\infty-\varepsilon}$ for all $N \geq N_0(\varepsilon)$. The statement is a liminf: the argument supplies no constant c with $D(N) \geq cN^{\alpha_\infty}$ for all N, and no such claim is made here.

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Corollary 2. *For prime q, let* $\mathrm{IM}(2,q)$ *be the largest induced matching in the point–line incidence graph of* $\mathbb{F}_q^2$*. Then for every* $\varepsilon > 0$*,*

$$\mathrm{IM}(2,q) \gg_\varepsilon q^{1/2+\alpha_\infty-\varepsilon}.$$

The corollary follows from Hunter–Pohoata–Verstraëte–Zhang [4, Prop. 2.3]: a square-difference-free $A \subseteq \{1,\dots,\lfloor q/10\rfloor\}$ yields an induced point–line matching in $\mathbb{F}_q^2$ of size $\Omega(\sqrt{q}\,|A|)$; the implication is a black box in A. The statement is for prime q only; it is not made for prime powers.

Context. The classical constructions are the height-1 case — a support with no square difference at all, no ranks, used as alternating digits — with exponent $\frac{1}{2}(1+\log t/\log m)$. The record for that quantity is $\frac{1}{2}(1+\log 12/\log 205) = 0.733412$, from $m = 205$, $t = 12$. For prime m the height-1 exponent never exceeds 3/4: a square-difference-free support mod p is an independent set in the Paley graph, of size at most $\sqrt{p}$ when $p \equiv 1 \pmod 4$, and is a single point when $p \equiv 3 \pmod 4$ or $p = 2$. Krachun [5] crossed 3/4 by making height, rather than width, the quantity that pays for itself in the CRT glue. This note keeps his glue and widens the supply of blocks: square-DAGs on square-free composite moduli, where height can be strictly smaller than width ($11 < 17$ and $12 < 19$ for the two certificates of Section 5).

2. Definitions

The arc set. For $m \geq 2$,

$$Q_m := \{\, z^2 \bmod m : z \in \mathbb{Z}\,\} \setminus \{0\},$$

the full image of squaring modulo m with 0 removed. For square-free m, by the Chinese remainder theorem, $d \in Q_m \cup \{0\}$ if and only if $d \bmod q$ is a square (possibly 0) in $\mathbb{F}_q$ for every prime $q \mid m$. Elements of Q_m may be *non-units* — zero in some CRT coordinates but not all. This is part of the definition, not an implementation choice: Remark 1 exhibits a counterexample showing that the unit-only variant makes Lemma A false. Sizes: $|Q_{235}| = 71$, $|Q_{299}| = 83$.

Square-DAGs and rankings. A *square-DAG* is a set $S \subseteq \mathbb{Z}/m\mathbb{Z}$ such that the digraph on S with arcs $x \to y$ whenever $y - x \in Q_m$ is acyclic. Acyclicity forces antisymmetry, since a two-way pair is a 2-cycle; no hypothesis on -1 being a nonresidue is needed.

A *ranking* of S is a map $h_0 : S \to \{0,\dots,H_0-1\}$ with $h_0(x) > h_0(y)$ for every arc $x \to y$; call H_0 the height of the ranking. The reversed longest-path rank is such a map, and it realizes the minimum H_0 over all rankings; that minimum value is the *height* of the DAG (the number of layers, equivalently the longest directed path counted in vertices).

Ranked blocks. A *ranked block* is a triple (C,h,H) with $C \subseteq \mathbb{Z}/P\mathbb{Z}$ and $h : C \to \{0,\dots,H-1\}$ strictly decreasing along every nonzero square difference mod P: if $x \neq y$ in C and $y - x$ is a square mod P, then $h(x) > h(y)$.

Paley chains. A *Paley chain* [5, Def. 2] for a prime $p \equiv 3 \pmod 4$ is a tuple $(s_0,\dots,s_{t-1})$ in $\mathbb{F}_p$ with $s_b - s_a$ a nonzero quadratic residue for all $a < b$. A Paley chain is a square-DAG whose digraph is a transitive tournament, so its height equals its length t.

Valuations. For a nonzero integer d, $v_m(d) := \max\{j : m^j \mid d\}$; for square-free $m = q_1 \cdots q_k$ this equals $\min_l v_{q_l}(d)$. For a nonzero class d in $\mathbb{Z}/m^{2e}\mathbb{Z}$, $v_m(d)$ is well defined and $< 2e$, computed on any representative.

3. The lifting lemma

The following lemma is the only new ingredient of this note.

Lemma A (composite even-digit lift)**.** *Let $m \geq 2$ be square-free, let S be a square-DAG on $\mathbb{Z}/m\mathbb{Z}$ with ranking h_0 of height H_0 and $|S| = t$, and let $e \geq 1$. Put*

$$C(m, S, e) := \Big\{ x = \sum_{j=0}^{2e-1} x_j m^j \;:\; 0 \leq x_j < m,\ x_{2j} \in S \text{ for } 0 \leq j \leq e-1 \Big\} \subseteq \mathbb{Z}/m^{2e}\mathbb{Z},$$

so that $|C(m, S, e)| = (mt)^e$, and

$$h(x) := \sum_{j=0}^{e-1} h_0(x_{2j})\, H_0^{e-1-j} \;\in\; \{0, \dots, H_0^e - 1\}.$$

Then for all $x \neq y$ in C with $y - x \equiv z^2 \pmod{m^{2e}}$ for some integer z and $y - x \neq 0$, one has $h(x) > h(y)$. That is, (C, h, H_0^e) is a ranked block on $\mathbb{Z}/m^{2e}\mathbb{Z}$ of size $(mt)^e$.

Proof. Step 0 (setup). Let $x \neq y$ in C and let $d := (y - x) \bmod m^{2e}$ satisfy $d \equiv z^2 \pmod{m^{2e}}$ with $d \neq 0$. Let r be the least base-m position at which the digit strings of x and y differ. Both are strings of $2e$ digits, so $r \leq 2e - 1$ automatically. As integers, $y - x = \sum_{l \geq r} (y_l - x_l) m^l$ with $y_r \neq x_r$, so $m^r \mid (y - x)$ and $m^{r+1} \nmid (y - x)$; the same holds for $d = (y - x) + \kappa m^{2e}$ with $\kappa \in \{0, 1\}$, since $r + 1 \leq 2e$ gives $m^{r+1} \mid m^{2e}$. Hence $m^r \mid d$, $m^{r+1} \nmid d$, that is $r = v_m(d)$. Step 2 sharpens this to r even and $r \leq 2e - 2$.

Step 1 (leading digit, no borrows). Every term with $l > r$ contributes a multiple of m to $(y - x)/m^r$, so $(y - x)/m^r \equiv y_r - x_r \pmod m$; and $d/m^r \equiv (y - x)/m^r \pmod{m^{2e-r}}$ because $\kappa m^{2e}/m^r \equiv 0$. The leading digit is therefore

$$\delta := (d/m^r) \bmod m \equiv y_r - x_r \pmod m, \qquad \delta \neq 0.$$

Step 2 (the valuation of a nonzero square residue is even, and $r \leq 2e - 2$). Fix a prime $q_l \mid m$ and set $v_l := v_{q_l}(z)$. Write $d = z^2 + \lambda m^{2e}$ for an integer λ. Then:

- if $2v_l < 2e$, then $v_{q_l}(z^2) = 2v_l < 2e \leq v_{q_l}(\lambda m^{2e})$, so $v_{q_l}(d) = 2v_l$, which is *even*; call such a coordinate *uncapped*;
- if $2v_l \geq 2e$, then $v_{q_l}(d) \geq 2e$ and no parity information is available; call such a coordinate *capped*.

Since $d \neq 0$ in $\mathbb{Z}/m^{2e}\mathbb{Z}$, not every coordinate can have $v_{q_l}(d) \geq 2e$, so at least one coordinate is uncapped, and

$$r = v_m(d) = \min_l v_{q_l}(d) = \min_{\text{uncapped } l} 2v_l,$$

a minimum of even numbers: the capped coordinates all have $v_{q_l}(d) \geq 2e$ and cannot achieve the minimum, which is $< 2e$. Hence $r = 2j$ is even with $0 \leq j \leq e - 1$.

Square-freeness is essential here. For $m = \prod q_l^{a_l}$ with a repeated prime factor, $v_m(d) = \min_l \lfloor v_{q_l}(d)/a_l \rfloor$ and evenness fails: take $m = 9$ and $d = 9 = 3^2$, where $v_9(d) = 1$ is odd.

Step 3 (global divisibility $m^j \mid z$). Every coordinate satisfies $v_{q_l}(z^2) \geq 2j$. For uncapped l this reads $2v_l \geq r = 2j$, since otherwise $v_{q_l}(d) = 2v_l < r$ would contradict r being the minimum; for capped l it holds because $2v_l \geq 2e \geq 2j + 2$. Hence $v_{q_l}(z) \geq j$ for every l, and by square-freeness $m^j \mid z$. Put $u := z/m^j \in \mathbb{Z}$.

Step 4 (the leading digit is a nonzero square mod m). From $d = z^2 + \lambda m^{2e} = m^{2j}(u^2 + \lambda m^{2e-2j})$ we get

$$d/m^{2j} \equiv u^2 \pmod{m^{2e-2j}}, \qquad \text{with } 2e - 2j \geq 2.$$

Reducing modulo m and combining with Step 1: $\delta \equiv u^2 \pmod m$ and $\delta \neq 0$, so $\delta \in Q_m$. Here u need not be a unit modulo m, so δ may be a non-unit square, zero in some CRT coordinates. This case occurs for genuine differences of the certificates of Section 5, which is why Q_m must be the full image of squaring.

Step 5 (rank drop). The first differing position $r = 2j$ is even, so $x_{2j}, y_{2j} \in S$, and $\delta \equiv y_{2j} - x_{2j} \pmod m$ with $\delta \in Q_m$ gives an arc $x_{2j} \to y_{2j}$, hence $h_0(x_{2j}) > h_0(y_{2j})$, a drop of at least 1. All even positions $2j' < 2j$ carry equal digits and contribute equally to h. Therefore

$$\begin{aligned} h(x) - h(y) &= \big[h_0(x_{2j}) - h_0(y_{2j})\big] H_0^{e-1-j} + \sum_{j'>j} \big[h_0(x_{2j'}) - h_0(y_{2j'})\big] H_0^{e-1-j'} \\ &\geq H_0^{e-1-j} - (H_0 - 1) \sum_{j'>j} H_0^{e-1-j'} = H_0^{e-1-j} - \big(H_0^{e-1-j} - 1\big) = 1 > 0. \quad \square \end{aligned}$$

Remark 1 (the unit-only variant is false). Suppose the arc set were taken to be the unit squares mod m only. Let $m = 15$, whose unit squares are $\{1, 4\}$. Then $S = \{0, 1, 6\}$ is acyclic for unit arcs — the only such arc is $0 \to 1$ — and $h_0(0) = 1$, $h_0(1) = 0$, $h_0(6) = 1$ is a valid ranking for that arc set. Lift with $e = 1$: $x = 0$ and $y = 36 = 6 + 2 \cdot 15$ both lie in C, and $y - x = 36 = 6^2$ is a square modulo 225 with $v_{15} = 0$ and leading digit 6, a non-unit square mod 15 (the full square set mod 15 is $\{1, 4, 6, 9, 10\}$). But $h(x) = 1 \leq 1 = h(y)$, so the lifted ranking fails, and Lemma B applied to it would manufacture a false SDF set. The definition Q_m = image of squaring minus 0 is therefore a hypothesis of the lemma.

Remark 2 (what differs from the prime case). Four points, and nothing else. (1) In Step 2, for prime p the capped case collapses, since a capped single coordinate forces $d \equiv 0$; for composite m a coordinate can hit the cap while the difference stays nonzero, so evenness must come from the minimum over uncapped coordinates. (2) In the prime case $p^j \mid z$ is immediate from $v_p(d) = 2j = 2v_p(z)$; for composite m the divisibility must be assembled coordinate by coordinate, the capped coordinates included. Step 3 is the composite form of the "$n^{2j} \mid z^2$, hence $n^j \mid z$" step inside Krachun's Lemma 4. (3) In Steps 4 and 5 the leading digit is automatically a unit square for prime p, and may be a non-unit square for composite m; transplanting the phrase "nonzero quadratic residue modulo p" verbatim would produce a false lemma. (4) Krachun extracts the order from his chain condition (2.1) — every forward difference along the chain a nonzero quadratic residue — together with -1 being a nonresidue mod p, which forces $b > a$; here that is replaced by "acyclicity yields a ranking," so $p \equiv 3 \pmod 4$ is not needed in any form, and the height t^e becomes H_0^e. Everything else — the digit representation, the borrow-free leading digit, the lexicographic descent, the size count — is Krachun's mechanism verbatim.

4. Krachun's lemmas, used as published

Lemma B (= [5, Lemma 4])**.** *Let $P = n^2$ be a perfect square, n any positive integer, with no primality or square-freeness hypothesis, and let (C, h, H) be a ranked block on $\mathbb{Z}/P\mathbb{Z}$. Then for every $L \geq 1$ there is a square-difference-free $A_L \subseteq \{1, \dots, (PH)^L\}$ with $|A_L| = |C|^L$.*

The mechanism, in one sentence: base-P words $(x_0, \dots, x_{L-1}) \in C^L$ give $X = \sum \bar{x}_j P^j$, where $\bar{x}_j \in \{0, \dots, P-1\}$ is the representative of x_j, with rank $h_L(X) = \sum h(x_j) H^{L-1-j}$, and $B_L = \{X + P^L h_L(X)\}$ works, because a positive square difference in B_L would force h_L not to decrease along its own reduction mod P^L, contradicting the strict decrease. For the proof see [5, Lemma 4]. Our $P = \prod m_i^{2e_i} = \left(\prod m_i^{e_i}\right)^2$ satisfies the hypothesis; nothing needs extending.

Lemma C (= [5, Lemma 5])**.** *For $1 \le i \le \ell$ let P_i be pairwise coprime perfect squares and (C_i, h_i, H_i) ranked blocks on $\mathbb{Z}/P_i\mathbb{Z}$. Under the CRT identification, $C = \prod C_i \subseteq \mathbb{Z}/P\mathbb{Z}$ with $P = \prod P_i$ and $h(x_1, \dots, x_\ell) = \sum h_i(x_i)$ is a ranked block of height $H = 1 + \sum(H_i - 1)$.*

The mechanism, in one sentence: the reduction of a square mod P is a square mod each P_i, possibly 0; a zero coordinate leaves its rank unchanged, a nonzero one drops it strictly, and at least one is nonzero. For the proof see [5, Lemma 5]. The statement never mentions chains or primes, so it accepts arbitrary ranked blocks as given.

5. The pool

Eleven blocks, all moduli pairwise coprime. Krachun's chain at $p = 23$ is omitted because $23 \mid 299$.

5.1. **Nine Paley chains.** These are Krachun's chains [5]; for these, $(m, t, H_0) = (p, t, t)$.

p	chain	t
3	$(0, 1)$	2
7	$(0, 4, 1)$	3
11	$(0, 3, 1, 4)$	4
19	$(0, 5, 11, 9, 16)$	5
31	$(0, 25, 14, 1, 19, 8, 2)$	7
43	$(0, 31, 9, 23, 4, 40, 1)$	7
59	$(0, 49, 15, 7, 16, 19, 35, 36, 5)$	9
71	$(0, 8, 12, 18, 48, 27, 1, 37, 20)$	9
103	$(0, 79, 25, 58, 55, 81, 4, 1, 34, 83, 59)$	11

Krachun's tenth chain, $23 : (0, 18, 1, 3, 4)$ with $t = 5$, is used here only to restate his constant $\alpha_\star$.

5.2. **Composite block 1:** $(m, t, H_0) = (235, 17, 11)$**, with** $235 = 5 \cdot 47$ **square-free.** Support with its exhibited ranking, written as *vertex: rank*:

$0:10$	$112:10$	$196:9$	$224:9$	$155:8$	$136:7$
$67:6$	$110:6$	$92:5$	$189:5$	$126:4$	$193:4$
$22:3$	$50:3$	$64:2$	$73:1$	$148:0$	

Seventeen vertices, ranks $0, \dots, 10$, so height 11. Checking this is a square-DAG with this ranking is one pass over the $17 \cdot 16$ ordered pairs: for each pair test whether the difference lies in Q_{235}, and if so that the rank strictly drops.

5.3. **Composite block 2:** $(m, t, H_0) = (299, 19, 12)$**, with** $299 = 13 \cdot 23$ **square-free.** Support:

$$7,\ 21,\ 25,\ 40,\ 46,\ 78,\ 83,\ 116,\ 153,\ 161,\ 165,\ 206,\ 207,\ 210,\ 212,\ 244,\ 264,\ 289,\ 292.$$

Nineteen vertices. The induced digraph under Q_{299} is acyclic, and its reversed longest-path rank has height 12; both are computed directly from the support in milliseconds.

Both blocks are presented as lower-bound certificates only. The cited heights $H_0 = 11$ and 12 are the heights of these exhibited certificates and nothing more; no minimality, maximality, or census claim is made about them or about the pool.

6. The exponent

Write m_i, t_i, H_i for the modulus, size, and height H_0 of pool block i; after Lemma A with multiplicity e_i, block i is a ranked block of height $H_i^{e_i}$, which is what Lemma C receives. Glue across the pool with Lemma C and feed the result to Lemma B at length L. Then

$$P = \prod_i m_i^{2e_i}, \qquad |C| = \prod_i (m_i t_i)^{e_i}, \qquad H = 1 + \sum_i \bigl(H_i^{e_i} - 1\bigr),$$

and the SDF sets produced by Lemma B have exponent

$$\alpha(e) = \frac{\sum_i e_i \log(m_i t_i)}{2\sum_i e_i \log m_i + \log\bigl(1 + \sum_i (H_i^{e_i} - 1)\bigr)}. \tag{1}$$

The limit. Every pool block has $H_i \geq 2$. Allocate $e_i(U) := \lfloor U/\log H_i \rfloor$ for a real parameter $U \geq \max_i \log H_i$, so that every $e_i(U) \geq 1$ and Lemma A applies. Then $e_i(U)\log H_i \in (U - \log H_i,\, U]$, so $e_i(U) = U/\log H_i + O(1)$, and

$$\max_i H_i^{e_i} \leq 1 + \sum_i \bigl(H_i^{e_i} - 1\bigr) \leq \ell \cdot \max_i H_i^{e_i} \quad \Longrightarrow \quad \log H = U + O(1),$$

with implied constants depending only on the pool. Substituting,

$$\text{numerator} = U \sum_i \frac{\log(m_i t_i)}{\log H_i} + O(1), \qquad \text{denominator} = U\Bigl(1 + 2\sum_i \frac{\log m_i}{\log H_i}\Bigr) + O(1),$$

both $\Theta(U)$, so $\alpha(e(U)) \to \alpha_\infty$ as $U \to \infty$, where

$$\alpha_\infty = \frac{\displaystyle\sum_i \frac{\log(m_i t_i)}{\log H_i}}{\displaystyle 1 + 2\sum_i \frac{\log m_i}{\log H_i}}. \tag{2}$$

This is Krachun's Theorem 1 computation with $\log t_i$ replaced by $\log H_i$; for a Paley chain $H = t$, so his formula is the special case.

Passage to all N. Fix $\rho < \alpha_\infty$ and choose U with $\alpha := \alpha(e(U)) > \rho$. Write $P = P(U)$, $H = H(U)$, $N_L := (PH)^L$, so Lemma B gives SDF $A_L \subseteq \{1, \dots, N_L\}$ with $|A_L| = |C|^L = N_L^\alpha$. For arbitrary $N \geq PH$ set $L := \lfloor \log N/\log(PH) \rfloor \geq 1$. Then $N_L \leq N$ and $A_L \subseteq \{1, \dots, N\}$ is square-difference-free, so

$$D(N) \geq |A_L| = N_L^\alpha > \bigl(N/(PH)\bigr)^\alpha = (PH)^{-\alpha} N^\alpha,$$

whence

$$\frac{\log D(N)}{\log N} \geq \alpha - \alpha\frac{\log(PH)}{\log N} \longrightarrow \alpha > \rho,$$

so $\liminf_{N\to\infty} \log D(N)/\log N \geq \rho$. Letting $\rho \nearrow \alpha_\infty$ proves Theorem 1. $\square$

Finite-stage certifiability. No limit is needed for an unconditional bound. Each finite U certifies its own exponent with an explicit constant: with $P = P(U)$ and $H = H(U)$,

$$D(N) \geq (PH)^{-\alpha(e(U))} N^{\alpha(e(U))} \qquad \text{for all } N \geq PH.$$

The limit is approached from below through these finite constructions: $\alpha(e(U)) = 0.744217$ at $U = 5$, 0.751542 at $U = 20$, 0.753210 at $U = 80$, 0.753616 at $U = 320$, gap of order $1/U$.

The value. Evaluating (2) on the eleven-block pool:

$$\alpha_\infty = 0.753741541837329405\ldots \qquad \text{(nearest double } 0.7537415418373294\text{)}.$$

For comparison, the same closed form over Krachun's ten Paley pairs (his (1.1)) gives $\alpha_\star = 0.752796455874514\ldots$

Scope. No optimality is claimed — not of the two certificates, not of their heights, not of the pool, and not of the method. The blocks are lower-bound certificates: exhibited objects that a reader checks in milliseconds. Krachun's §3 discussion of the limit of the approach is heuristic, restricted to prime moduli, and stated by its author to be without precise proofs, so it bears on none of the above.

7. Verification and formalization

The finite content of the note — the arc sets Q_{235} and Q_{299} rebuilt from the definition, both certificates checked for size, acyclicity, and longest-path height, all ten Paley chains checked for primality, $p \equiv 3 \pmod 4$, distinctness, and every forward difference a nonzero quadratic residue, and both constants recomputed in high-precision decimal — is re-checked by the ancillary script `verify.py` (Python 3, standard library only, well under a minute, all data inlined, nonzero exit status on any failure). The flag `--lift-demo` additionally builds the $e = 1$, $L = 1$ lifted integer set for the 235 block and brute-checks square-difference-freeness — an empirical witness of Lemmas A and B at the smallest scale, a demonstration, not a proof. The script re-checks data stated in full above; it is not an input to the mathematics.

The complete argument is moreover formalized in Lean 4 [6] with Mathlib [8]: the definitions of Section 2, Lemma A, Lemmas B and C reproved rather than cited, the exponent computation, and Theorem 1 in both liminf and pointwise ε-form, together with the numerical statement $0.7537 < \alpha_\infty$, certified by kernel-checked integer comparisons with no floating-point or transcendental arithmetic anywhere. The development comprises 126 declarations, every one depending only on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`). A source snapshot of the formalization accompanies this submission as ancillary files (`anc/lean/`), with the toolchain and dependency versions pinned by `lean-toolchain` and `lake-manifest.json`, so the kernel check can be reproduced against the exact versions used. The formalization is maintained at `https://github.com/JD-Jones-ASES/fs-lower-bound-lean`; the note's repository, including the verification script, is at `https://github.com/JD-Jones-ASES/fs-lower-bound`.

Provenance

This note reports AI-generated mathematics with a human managing the workflow. The construction, proofs, and text were produced by Claude Code (Fable 5, Anthropic), with contributions from Codex (GPT 5.6 Sol, OpenAI) and Grok Build (Grok 4.6, xAI); the author directed the work and takes responsibility for the content. Verification does not require trusting any of these systems: everything needed to reimplement the construction is stated above, the finite data are re-checked by the ancillary script, and the complete argument is machine-checked in Lean 4 (Section 7).

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